2313666 殷家兴 自动化 自检②

2.2 证明

$U = E$

当应变片不受力时，$\Delta R = 0$ 且此时桥路处于平衡状态

因此 $R_1 = R_4$ ，设初始平衡状态下 $R_1 = R_4 = R_0$

则 R_1 受拉力时 $R_1 = R_0 + \Delta R$ 　 R_4 受压力时 $R_4 = R_0 + \Delta R$

$$U_{cd} = \frac{R_1}{R_1+R_2}U - \frac{R_4}{R_4+R_3}U = \frac{R_1(R_4+R_3) - R_4(R_1+R_2)}{(R_1+R_2)(R_4+R_3)}U$$

$$= \frac{(R_0+\Delta R)(R+R_0-\Delta R) - (R_0-\Delta R)(R+R_0+\Delta R)}{(R+R_0+\Delta R)(R+R_0-\Delta R)}U$$

$$= \frac{2R\Delta R}{(R+R_0)^2 - \Delta R^2}U = \frac{2R\Delta R}{(R+R_0)^2-\Delta R^2}E$$

取 $R_0 = R$ ，设 $\Delta R^2 \ll R^2$，上式中 ΔR^2 项可省略，化为：

$$U_{cd} = \frac{2R\Delta R}{4R^2}E = +\left(\frac{E}{2}\right)\left(\frac{\Delta R}{R}\right)$$

2.4 解：

$m = \dfrac{R_{max}}{R_L}$ 　 当 $m < 0.1$ 时 　 $R_L > 10R_{max}$

$$\begin{cases} \delta_2 = 1\pm\left[1 - \dfrac{1}{1+m'r(1-r)}\right] \cancel{\times 100\%} < 0.1 \\ r = \gamma = \dfrac{R_x}{R_{max}} = \dfrac{1}{2} \qquad m' = \dfrac{R_{max}}{R_L} \end{cases}$$

解得：$m' < \dfrac{4}{9}$ 即 $\dfrac{R_{max}}{R_L} < \dfrac{4}{9}$ 　 $\cancel{R_{max} < \frac{4}{9}R_L}$ 　 $R_L > \dfrac{9}{4}R_{max}$

2.6 解： 令 $R_x = xR_{max}$

当 $R_F = 50\Omega$ 时

$m_1 = \dfrac{R_p}{R_F} = 2$ 　 $r_x = \dfrac{R_x}{R_{max}} = x$

$$y = \frac{r_x}{1+r_x m_1(1-r_x)} = \frac{x}{1+2x(1-x)} \qquad x \in \{0, 0.1, 0.2, \ldots, 1.0\}$$

x	0	0.1	0.2	0.3	0.4	0.5	0.6	0.7	0.8	0.9	1.0
y	0	$\frac{5}{59}$	$\frac{5}{33}$	$\frac{15}{71}$	$\cancel{\frac{10}{49}}$ $\frac{10}{37}$	$\frac{1}{3}$	$\frac{15}{37}$	$\frac{35}{71}$	$\frac{20}{33}$	$\frac{45}{59}$	1

当 $R_F = 500\Omega$ 时 $\quad m_2 = \dfrac{R_p}{R_F} = \dfrac{1}{5} = 0.2$

$$y = \frac{r_x}{1 + m r_x (1 - r_x)} = \frac{x}{1 + 0.2x(1-x)} \qquad x \in \{0, 0.1, 0.2, \cdots, 1.0\}$$

x	0	0.1	0.2	0.3	0.4	0.5	0.6	0.7	0.8	0.9	1.0
y	0	$\frac{50}{509}$	$\frac{25}{129}$	$\frac{150}{521}$	$\frac{50}{131}$	$\frac{10}{21}$	$\frac{75}{131}$	$\frac{350}{521}$	$\frac{100}{129}$	$\frac{450}{509}$	1

2.10 解:

$$\varepsilon_x = \frac{6(l-x)}{W E t^2} F = \frac{6 \times (0.25 - \frac{0.25}{2})}{0.06 \times 70 \times 10^5 \times (3 \times 10^{-3})^2} \times 0.5 \approx 0.1$$

$$\frac{\Delta R}{R_0} = \cancel{K_0 \varepsilon}\ K\varepsilon = 2.1 \times 0.1 = 0.21$$

$$\Delta R = 0.21 R_0 = 25.2\Omega$$

$$R_1 = R_3 = R_0 + \Delta R = 145.2\Omega$$

$$R_2 = R_4 = R_0 - \Delta R = 94.8\Omega$$